could not support more than a few calls per second, and your management's projection is in the hundreds of emails written per second (they are very optimistic, and a bit reckless, but let's not worry about that for now!). Your first instinct would be to ask for a remote distributed cache, like Memcached, Cassandra or Redis. A roundtrip to the cache server alone would take in the range of a hundred milliseconds, best case scenario, which would be good. But neither spinning up a new server for the cache, nor buying it as a cloud service could be feasible for you, budget-wise.

In the end, you decide you only have one way to solve this: asynchronously get your contacts list when you log in (or, more lazily, the first time you click on compose during current browser session), save the contacts list in your web page's session storage space, and check the local copy of that data every time you look for existing contacts.

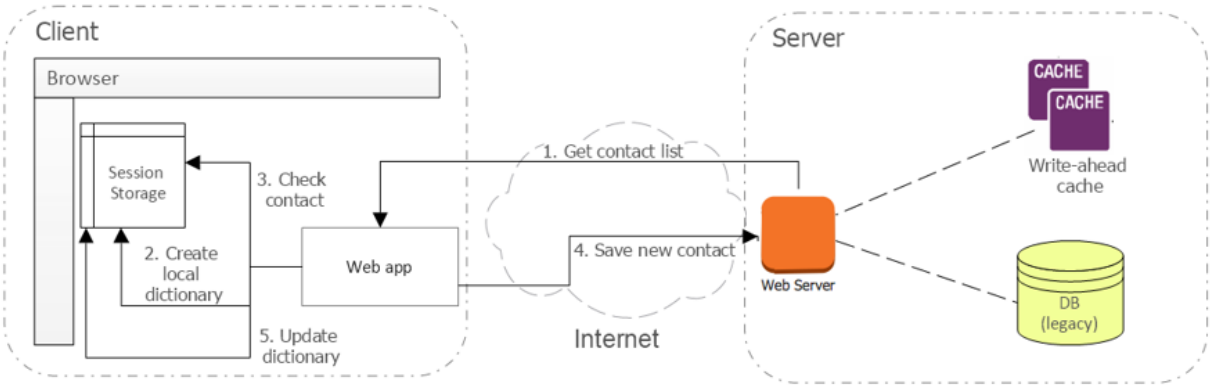


Figure 4.2 A possible architecture for the "save new contact (with suggestion)" feature. On the client side, the web app receives the contacts list from the web server, and with those data creates a dictionary on the session storage. Whenever the user adds a recipient on an email, the web app checks the